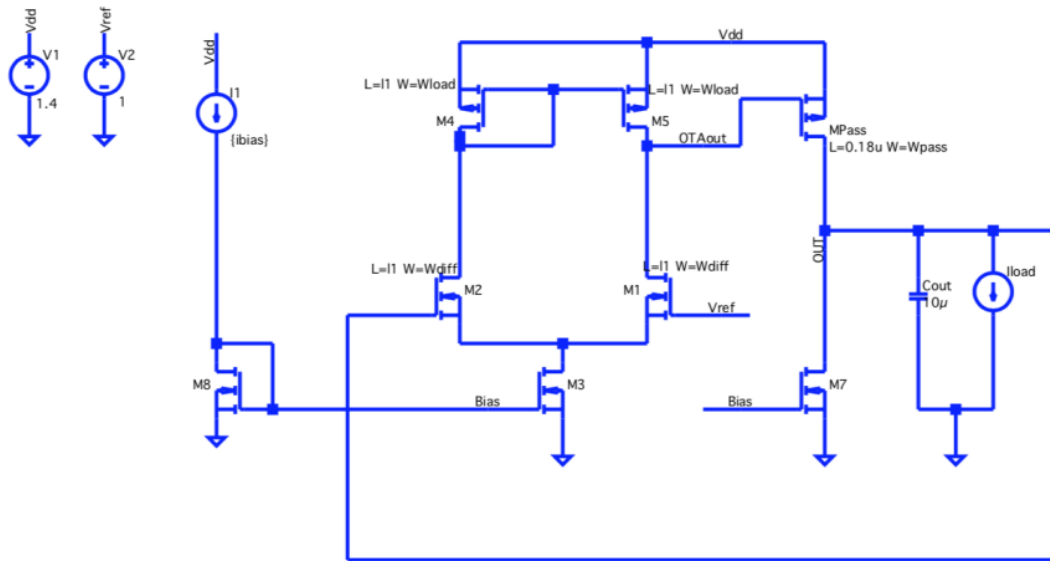




Schematic

Module 2:

1. What is the drop-out voltage?

Answer: $V_{DD} - V_{OUT} = 0.4 \text{ V}$

2. What is the target DC loop gain, assuming we add a 20 dB margin to DC PSRR?

Answer: $\text{DC PSRR} = 40 + 20 \text{ dB} = 60 \text{ dB}$. $\text{DC Loop Gain} = 1/\text{DC PSRR} = 60 \text{ dB (approx.)}$

3. What is the mV peak-to-peak spread for Vout for load transients?

Answer: $V_{OUT} = 1 \text{ V}$. V_{OUT} variation due to load transient = 1% (0.01V).
Therefore, mV peak to peak spread = 990mV to 1010mV.

4. Which pole will be the dominant one - the miller (Pole at OTAout) or the output pole, the pole at OUT? Why?

Answer: Pole at OTAout = $1 / ((r_{ds,p} \parallel r_{ds,n}) * (C_{gg,pass} + C_{gd,p} + C_{gd,n}))$.

Pole at OTAout $\approx 1 / ((r_{out,diff}) * C_{gg,pass})$

Output Pole (at OUT) = $1/((r_{ds,pass} || R_{Bias}) * (C_{OUT} + C_{gd,bias} + C_{gd,pass}))$.

Output Pole (at OUT) $\approx 1/(r_{ds,pass} * C_{OUT})$.

Now, $r_{out,diff}$ is in the range of 10^6 (MOhm) and $C_{gg,pass}$ is in the range of 10^{-9} (nF). Therefore, **Pole at OTAout is in 10^3 (KOhm).**

$C_{OUT} = 10 \mu F$ ($10 * 10^{-6} F$)

$r_{ds,pass}$ is in the range of 10^6 Ohm.

Therefore, **output pole (at OUT) is in the range of 10^0 Ohm.**

Pole at VOUT is dominant.

Module 3:

PMOS LDO Design:

PASSFET: PMOS

Differential Amplifier: NMOS I/P Pair

Module 4:

PASSFET Sizing:

pGm/ID (Vov = 0.2 V)	10
Vds(V)	0.4
Iload(A)	100m
L,min(M)	180 n
pId/W (A/m)	9.328
Wpass(M)	10.72m
gmro(passfet)	32.22 V/V
Target Open Loop Gain	1000 V/V
A,diff	31.036 V/V
Gm (S)((gm/Id) * Id)	1
ro,pass (Ohm)	32.22
Gm/2pi*Cgg	5.775 G
Cgg,pass (F) = (Gm/(Gm/2pi*Cgg))	27.54p

$$wp2 = 1 / (rout,diff * Cgg,pass)$$

$$wp1 = 1 / (ro,pass * COUT)$$

$$fp1 = 1 / (2pi * 32.22 * 10 * 10^{-6}) = 494.213Hz$$

Module 5:

STABILITY ANALYSIS

1. What is the dominant pole expression (the pole at Output) and estimated value (f_{p1})?

Answer:

$$\text{Pole at Output}(wp1) = 1/((ro,pass||Rbias)*(COUT+Cgd,Bias+Cgd,pass))$$

$$ro,pass \gg Rbias$$

$$COUT \gg Cgd,bias, Cgd,pass$$

$$COUT = 10\mu F \text{ (given)}$$

$$fp1 = 1/(ro,pass * COUT * 2\pi) = 1/(2\pi * 32.22 * 10 * 10^{-6}) = 494.213\text{Hz}$$

2. What is the low-frequency loop gain expression (A) and estimated value?

Answer: **Low Frequency Loop Gain = $A_{diff} * A_{pass}$**

$$A_{pass} = gm_{pass} * ro_{pass}$$

$$A_{diff} = gm_{diff} * (rds,p||rds,n)$$

$$\text{Targeted Open loop gain} = 1000 \text{ V/V}$$

$$\text{DC Loop Gain} = gm_{diff} * (rds,p||rds,n) * gm_{pass} * ro_{pass} = 1000 \text{ V/V} \approx 60 \text{ dB.}$$

3. What is the expected unity gain bandwidth (ω_u)?

$$\text{Answer: Unity Gain Frequency}(wu) = \text{Loop Gain}(A) * Fp1$$

$$\text{Unity Gain Frequency} = 1000 \text{ V/V} * 494.213 \text{ Hz} = 0.494 \text{ MHz.}$$

4. How do f_{p1} , A and ω_u vary with I_{load} - Please comment.

Answer:

$I_{load} \text{ (A)}$	$Fp1 \text{ (Hz)}$	$A \text{ (V/V)}$	$Wu \text{ (Hz)}$
10m	49.4213	1000	49421
100m	494.213	1000	0.494M

Wu for heavy load is higher than light load.

5. How would you position the second pole- at OTAout? What would be its value for a worst-case phase margin of 45 degrees?

Answer: Pole at OTAout (ω_{p2}) = $1/((r_{ds,n} || r_{ds,p}) * C_{gg,pass})$

For Phase Margin = 45 Degree,

$$\omega_{p2} = \omega_{u(Heavy)}/2 * \pi = \text{loop gain} * \omega_{p1} = 0.494 \text{M Hz.}$$

6. What is the expression of the pole at OTAout?

Answer: Pole at OTAout (ω_{p2}) = $1/((r_{ds,n} || r_{ds,p}) * C_{gg,pass})$

7. What is the value of the output impedance of the first stage- the differential amplifier?

Answer:

For Heavy Load,

$$\omega_u = \omega_{p2} = 0.494 \text{M Hz}$$

$$1/(2\pi * r_{out,diff} * C_{gg,pass}) = 0.494 \text{M Hz}$$

$$R_{out,diff} = 1/(2\pi * 0.494 \text{M} * 27.54 \text{p}) = 11704.408 \text{ Ohm}$$

Module 6:

Differential Amplifier FET Sizing:

Differential I/P FET:

1. What would be the target value of low-frequency gain of stage 1 $gm_{ndiff} * r_{ondiff} || r_{opload}$?

Answer: Target value of DC PSRR = 60 dB = 1000 V/V

Therefore, Loop Gain = 60 dB = 1000 V/V

From Techplot, $L_{min} = 180$ nm, $gm/Id = 10$, $gmro$ (PASSFET) = 32.22 V/V.

$$A_{pass} * A_{diff} = 1000 \text{ V/V}$$

$$A_{diff} = 1000/A_{pass} = 1000/32.22 = 31.036 \text{ V/V.}$$

The target value of low frequency gain of Differential Amplifier = 31.036 V/V (29.83 dB).

2. What is the output impedance of the NMOS diff pair or PMOS Load, assuming $r_{ondiff} = r_{opload}$?

Answer: $A_{diff} = 31.036 \text{ V/V}$

Assuming $r_{ondiff} = r_{opload}$,

$$A_{diff} = gm_2(ro_1 || ro_5) = gm_2 * (ro_1/2) = 31.036 \text{ V/V}$$

$$R_{out,diff} = ro_1/2 = 11704.408 \text{ Ohm}$$

$$r_{ondiff} = r_{opload} = 23408.8 \text{ Ohm}$$

Therefore, the output impedance of the NMOS diff pair or PMOS Load = 23408.8 Ohm.

3. What is the target gm of the NMOS diff pair?

Answer: $A_{diff} = 31.036 \text{ V/V}$

Output Impedence of Differential Amplifier = $ro_1/2 = 11704.408 \text{ Ohm.}$

gm of NMOS diff pair = 2.651 mS

4. You may use the entire quiescent current budget for the differential amplifier, each branch gets $I_q/2(=250 \text{ uA})$. Given this information, what would be the gm/I_d for NMOS diff pair?

Answer: gm/I_d of NMOS diff = $gm / (I_q/2) = 2.65\text{m} / 250\text{u} = 10.606$.

5. What is the length of the NMOS input pair? Use the techplots for $gmro$ vs. ngm_id .

Answer: $gmro = 62.072 \text{ V/V}$

$gm/I_d = 10.606$

From Techplot, Length of NMOS I/P = 540 nm.

6. Use $nidw$ vs. ngm_id to evaluate I_d/W and then the Width of the NMOS diff pair.

Answer: From Techplot, for $gm/I_d = 10.606$ and $L = 540\text{nm}$, $I_d/W = 7.795$.

$I_q = 250\text{u A}$

Therefore, the width of the NMOS diff pair $W = 32.0718\text{um}$.

PMOS Load Sizing:

7. Assuming gm/I_d of FETs is the same as PMOS pass transistor can help increase the PSRR by reducing V_{ds} mismatch. So, what is the gm/I_d of the PMOS load for the Differential Amplifier?

Answer: gm/I_d of the PMOS load for the Differential Amplifier = 10.

8. Assuming the Current through each PMOS load is $250 \text{ }\mu\text{A}$, that is the gm ?

Answer: $gm = (gm/I_d) * I_d = 10 * 250\text{u S} = 2500 \text{ uS}$.

9. Now that you know gm and ro , find the device lengths and Width from techplots - $gmro$ vs. gm/I_d and I_d/W vs. gm/I_d .

Answer: $gm = 2500 \text{ uS}$, $ro = 23408.8 \text{ Ohm}$.

$gmro = 58.522 \text{ V/V}$

$gm/I_d = 10$

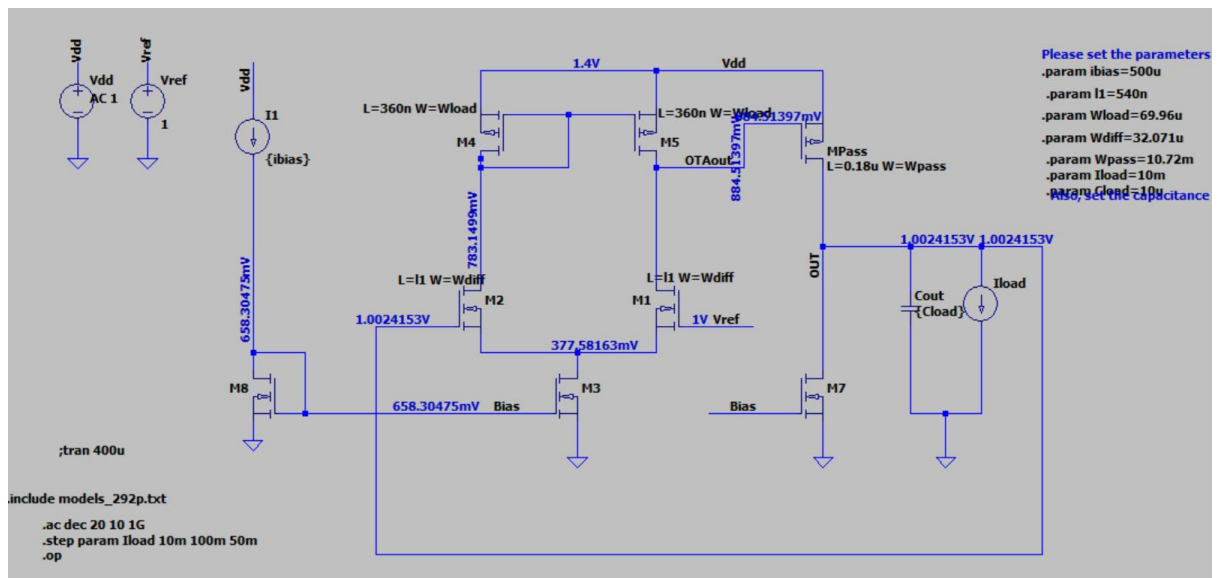
From Techplot, $L = 360 \text{ nm}$.

$I_d/W = 3.573$. $W = 250\text{u}/3.573 = 69.969 \text{ um}$. $W = 69.969 \text{ um}$.

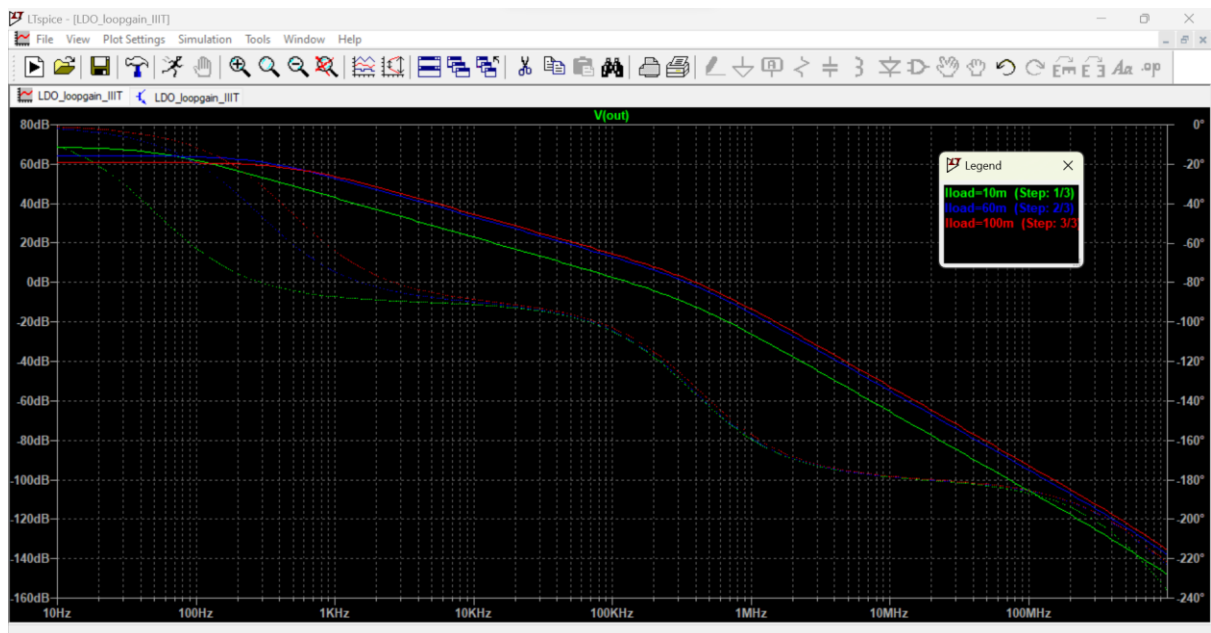
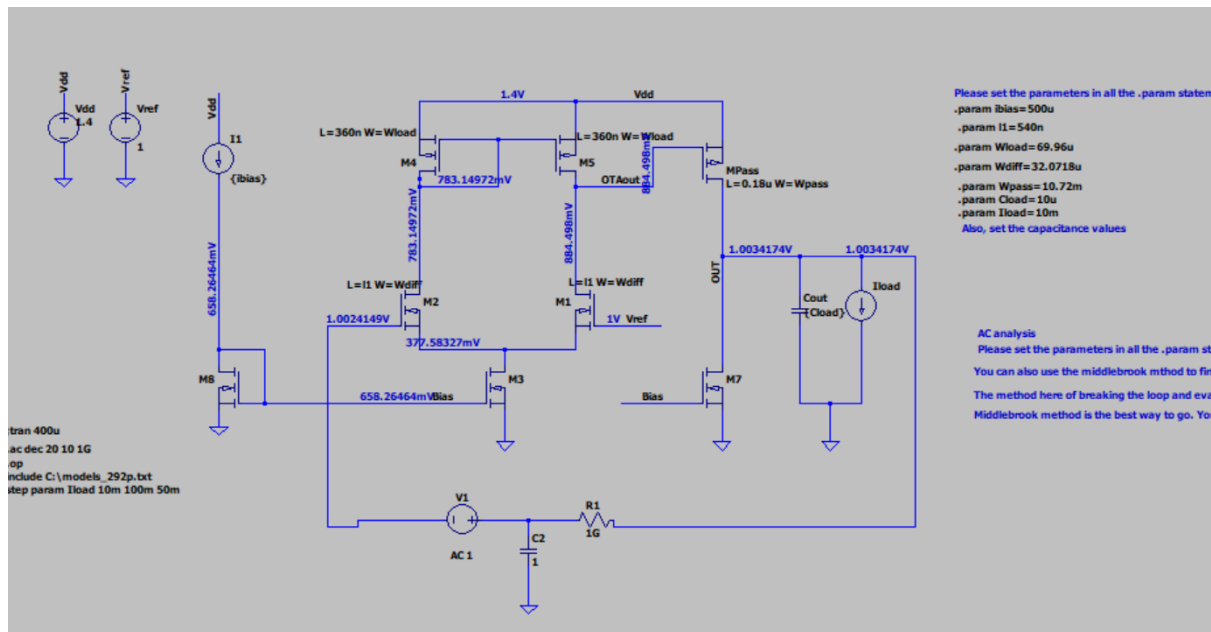
MOSFET	I(A)	Gm/Id	Width (M)	Length (M)
PASSFET	100m	10	10.72m	180n
Diff I/P (NMOS)	250u	10.606	32.0718u	540n
PMOS Load	250u	10	69.969u	360n
M8, M3, M7	500u	10	18.228u	180n

Module 7:

DC Operating Point Simulation:

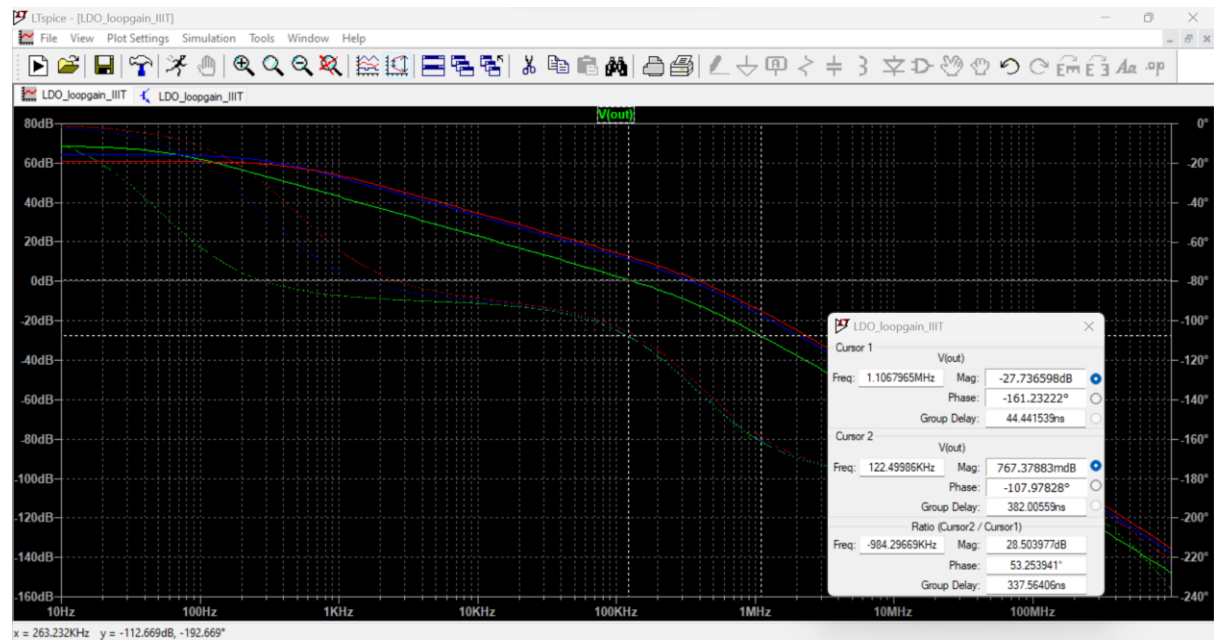


Checking the Loop gain and Phase Margin at different loads:

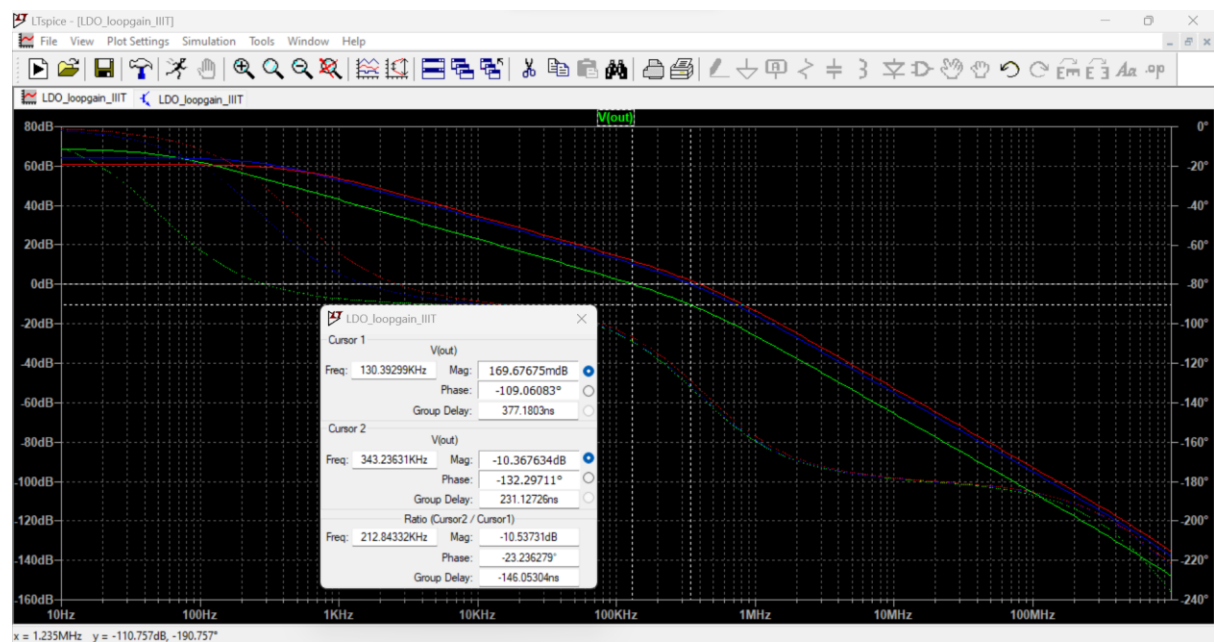


Bode Plot for different loads

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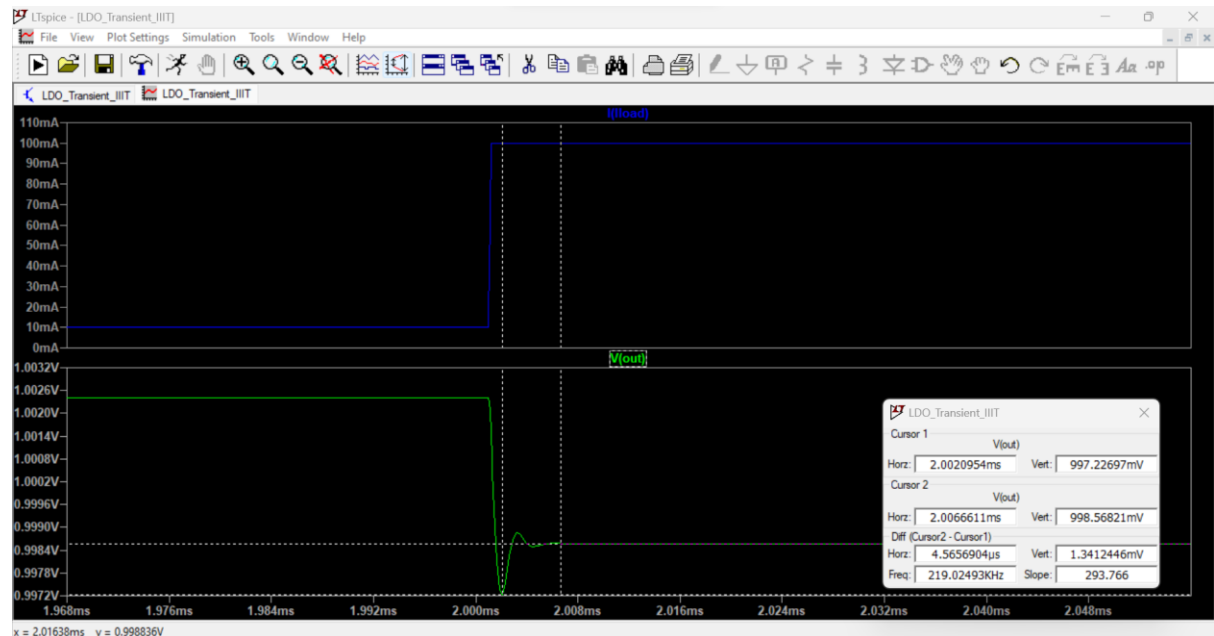
Phase Margin observed for light load = 72.03 degree.



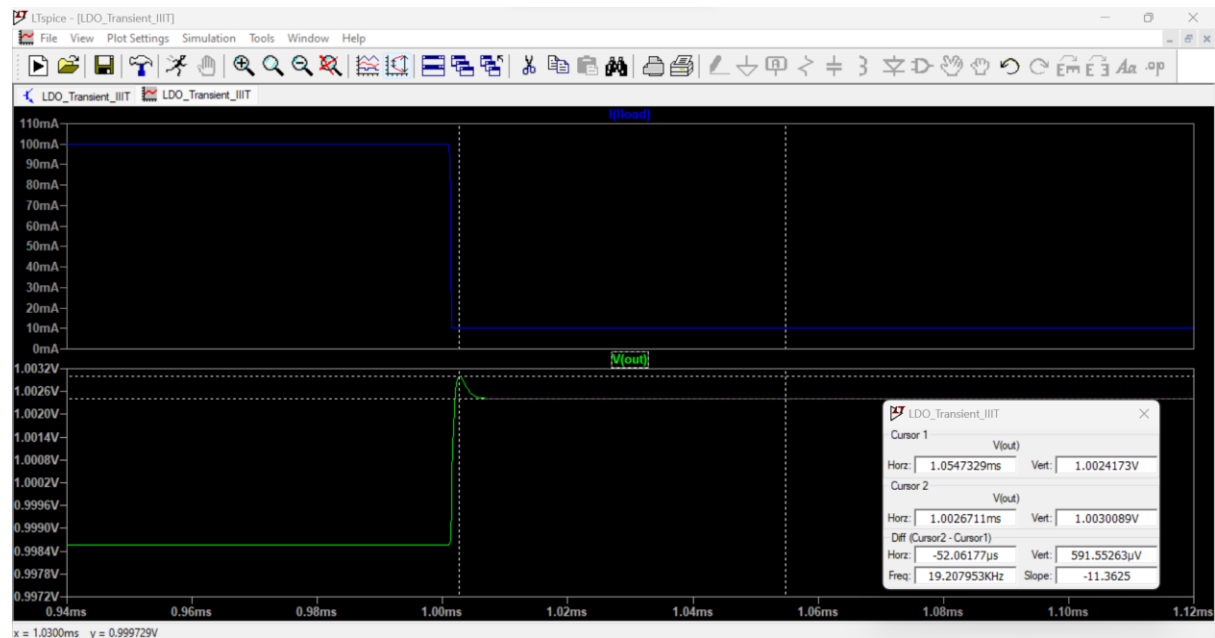
Phase Margin observed for heavy load = 47.71 degree.

Checking Overshoot and Undershoot:

As per the specification, the mV spread due to load transient = 990 mV to 1010mV.



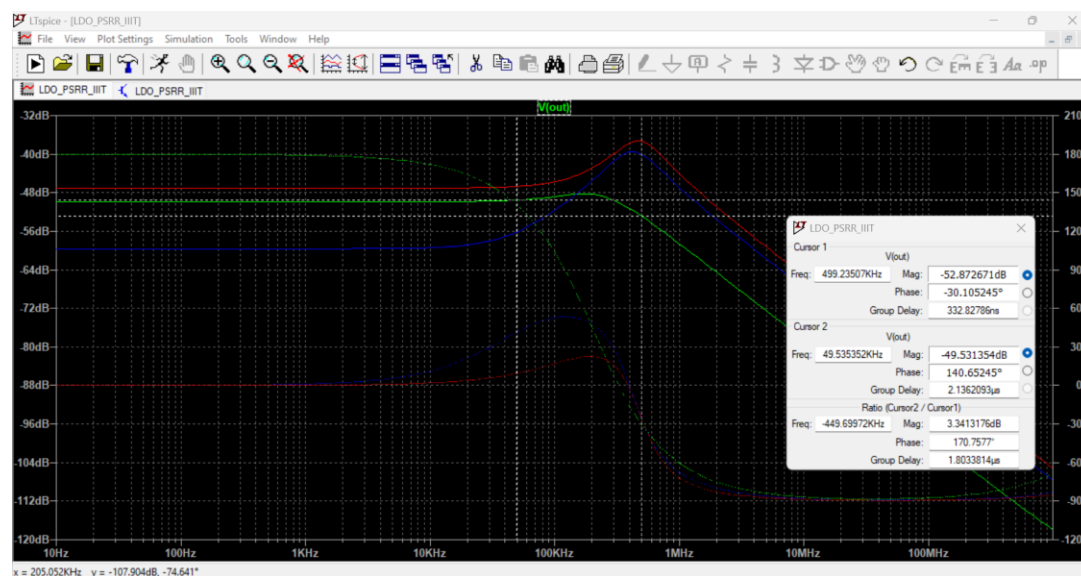
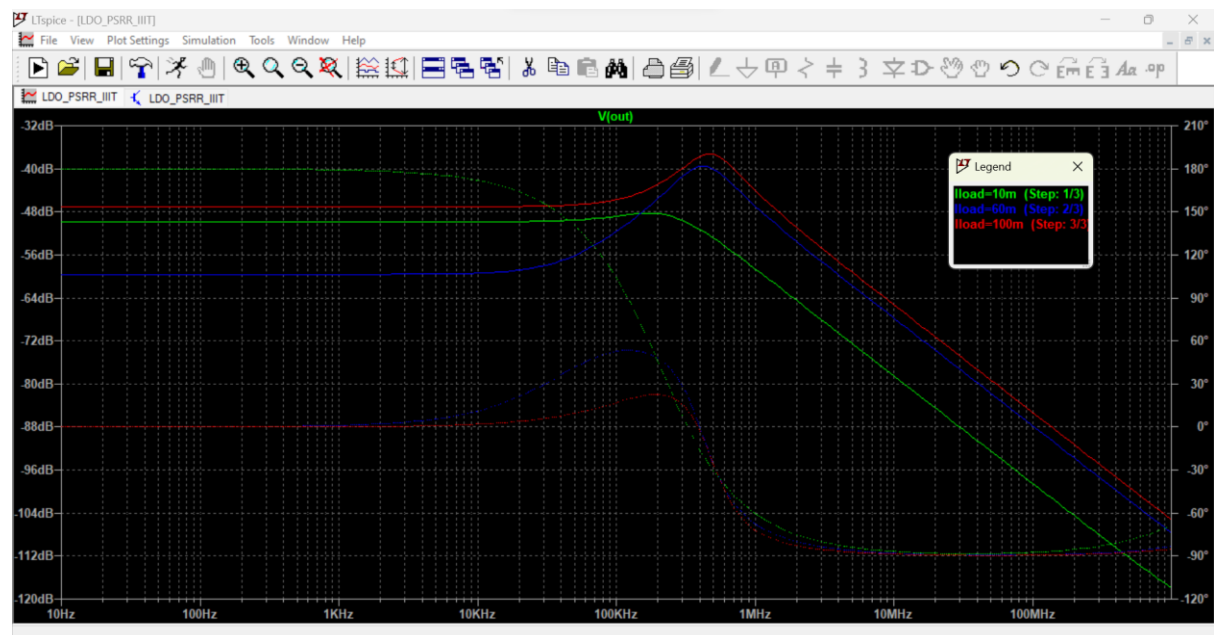
According to the simulation, the undershoot touches 997.2mV.



Peak Overshoot reaches 1003.0089 mV.

Therefore, the Vout spread due to load transient is within the range.

PSRR curve:



From the simulation,

Iload (A)	Fu (Hz)	Fp2(Hz)	Low Freq. PSRR (dB)	High Freq. PSRR (dB)
10m	49.1K	0.491M	-50.056 (at 10 Hz)	-58.625 (at 1MHz)
100m	0.491M	0.491M	-47.25 (at 10 Hz)	-44.27 (at 1MHz)

PSRR is comparable to the worst-case Low Frequency and High Frequency ranges given in the specification.